

NIZHYN, Ukraine

WMO: 332460

Lat: 51.0381N

Lon: 31.9003E

Elev: 413

StdP: 14.48

Time Zone: 2.00 (EUE)

Period: 94-16

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -2.0	(c) 3.0	(d) -6.8	(e) 3.9	(f) -0.5	(g) -2.5	(h) 4.9	(i) 3.8	(j) 16.2	(k) 28.9	(l) 14.2	(m) 27.2	(n) 2.8	(o) 40	(p) 0.476

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 18.6	(c) 86.2	(d) 69.2	(e) 83.4	(f) 67.7	(g) 80.8	(h) 66.5	(i) 71.7	(j) 82.4	(k) 69.7	(l) 80.2	(m) 68.0	(n) 77.8	(o) 3.7	(p) 70

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
67.6	103.2	78.0	65.8	96.5	75.4	64.3	91.5	73.4	35.7	82.4	33.9	80.6	32.6	77.9	81.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 13.9	(b) 11.9	(c) 10.1	(e) -8.6	(f) 90.9	(g) 6.7	(h) 3.0	(i) -13.4	(j) 93.1	(k) -17.4	(l) 94.9	(m) -21.1	(n) 96.5	(o) -26.0	(p) 98.7		
(a) 13.9	(b) 11.9	(c) 10.1	(e) -9.0	(f) 74.9	(g) 6.7	(h) 3.3	(i) -13.8	(j) 77.3	(k) -17.7	(l) 79.2	(m) -21.4	(n) 81.1	(o) -26.3	(p) 83.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	46.1	23.7	27.6	35.1	48.2	58.7	64.4	68.4	65.2	55.8	44.7	35.4	24.6	(6)
(7)		DBStd	17.78	10.13	9.76	7.76	8.01	7.73	6.12	5.05	5.61	7.07	7.89	9.05	10.82	(7)
(8)		HDD50	3507	815	628	465	128	16	0	0	0	25	202	441	787	(8)
(9)		HDD65	7221	1280	1048	928	506	224	87	24	68	286	629	889	1252	(9)
(10)		CDD50	2073	0	0	2	72	285	433	571	470	200	38	2	0	(10)
(11)		CDD65	312	0	0	0	1	28	70	130	73	10	0	0	0	(11)
(12)		CDH74	2968	0	0	0	6	274	606	1146	801	130	5	0	0	(12)
(13)		CDH80	720	0	0	0	0	39	124	318	214	25	0	0	0	(13)
(14)	Wind	WSAvg	4.2	4.9	5.1	5.3	4.5	4.1	3.7	3.0	3.0	3.5	3.8	4.7	4.8	(14)
(15)	Precipitation	PrecAvg	24.0	1.5	1.3	1.4	1.9	2.9	2.6	2.9	2.2	2.3	1.8	1.9	1.5	(15)
(16)		PrecMax	29.4	2.9	2.4	3.0	4.3	5.8	3.9	7.6	4.9	4.9	4.9	4.8	3.0	(16)
(17)		PrecMin	15.4	0.5	0.4	0.1	0.3	1.4	0.9	0.7	0.0	0.3	0.1	0.1	0.4	(17)
(18)		PrecStd	3.8	0.6	0.6	0.7	1.1	1.3	1.0	1.8	1.4	1.4	1.2	1.1	0.7	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	42.3	47.1	61.0	74.1	83.5	86.6	89.9	89.9	82.9	72.5	56.6	45.9	(19)
(20)			MCWB	40.9	43.0	47.9	58.5	64.8	68.9	71.6	69.8	64.7	61.6	52.0	43.4	(20)
(21)		2%	DB	39.6	43.3	55.0	69.8	80.6	83.7	86.5	85.6	77.8	65.7	52.5	42.6	(21)
(22)			MCWB	37.3	40.4	45.7	55.6	64.4	67.8	70.2	68.8	62.8	57.6	49.5	41.1	(22)
(23)		5%	DB	37.0	40.6	50.6	65.9	77.1	80.7	83.7	81.7	72.8	60.3	49.7	39.9	(23)
(24)			MCWB	35.3	38.2	43.4	53.9	63.8	66.8	68.9	66.5	60.6	54.7	47.7	38.7	(24)
(25)		10%	DB	35.0	38.3	46.3	62.3	73.2	77.4	80.8	78.3	68.3	56.4	46.9	36.7	(25)
(26)			MCWB	33.7	36.4	40.5	52.0	61.2	65.1	67.8	64.5	58.7	52.0	45.0	35.2	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	41.3	44.2	50.0	60.2	69.5	72.0	73.8	73.3	67.1	63.1	53.0	44.1	(27)
(28)			MCD B	41.7	46.5	59.0	70.7	77.3	82.2	85.4	85.7	80.0	70.3	56.1	45.2	(28)
(29)		2%	WB	37.6	40.8	46.8	57.4	66.6	69.9	72.2	70.0	64.5	58.5	50.1	41.2	(29)
(30)			MCD B	38.8	42.7	53.8	66.4	76.7	80.5	83.5	81.9	74.3	63.9	51.7	42.4	(30)
(31)		5%	WB	35.5	38.6	44.0	55.2	64.4	68.1	70.5	68.1	62.0	55.4	47.9	38.8	(31)
(32)			MCD B	36.6	40.4	49.4	63.0	73.4	78.0	80.9	78.5	70.4	60.1	49.6	39.9	(32)
(33)		10%	WB	33.8	36.6	41.5	53.0	62.4	66.1	68.7	66.4	59.8	52.5	45.1	35.5	(33)
(34)			MCD B	34.8	38.1	45.4	60.4	71.9	75.9	78.1	75.8	66.0	55.8	46.9	36.5	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	8.1	9.6	12.4	17.2	19.8	18.1	18.6	20.6	17.7	15.0	8.1	8.3	(35)
(36)			MCDBR	6.7	10.2	19.3	23.3	24.0	23.3	23.7	26.5	26.2	21.8	10.9	8.2	(36)
(37)			MCWBR	5.8	8.4	11.8	11.6	11.2	11.3	10.5	11.4	12.8	13.5	8.5	7.5	(37)
(38)		5% WB	MCDBR	6.1	9.6	17.5	19.7	21.0	21.1	21.3	22.8	22.1	19.7	9.8	8.0	(38)
(39)			MCWBR	5.6	8.6	11.6	10.9	11.2	11.8	10.4	10.8	11.9	13.4	8.1	7.6	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.309	0.341	0.372	0.438	0.409	0.397	0.429	0.431	0.399	0.363	0.344	0.313	(40)	
(41)		taud	2.171	2.103	2.166	2.103	2.267	2.322	2.246	2.225	2.276	2.347	2.274	2.216	(41)	
(42)		Ebn at Noon	216	240	258	253	269	273	261	253	247	235	203	195	(42)	
(43)		Edh at Noon	24	33	38	45	40	39	41	40	34	26	22	20	(43)	
(44)	All-Sky Solar Radiation	RadAvg	270	539	953	1361	1796	1901	1834	1598	1079	598	240	175	(44)	
(45)		RadStd	49	89	149	131	145	154	120	113	143	97	39	43	(45)	

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (10 neighbors)	+1.26	N/A	N/A	+1.82	+0.79	N/A	-236	-340	+198	+94

Nomenclature: See separate page